

Model-Based and Data-Driven Control of a Self-Balancing Robot Under Dynamic Conditions using Dissipativity

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Abstract—This project proposes a hybrid control workflow for a two-wheeled self-balancing robot under dynamic operating conditions. First, a model-based controller is designed from the linearized state-space model and simulated in MATLAB to establish a stabilizing baseline. The controlled system is then used to generate input-state trajectory data, for direct data-driven controller synthesis. To improve robustness, a dissipativity-based formulation is introduced as an additional LMI-based performance constraint. Rather than modifying the data equation, dissipativity is used to bound the disturbance-to-performance-output energy gain through a disturbance channel and selected performance output. MATLAB simulations compare the model-based, data-driven, and dissipativity-enhanced controllers under initial angle deviations, disturbance inputs, and parameter variations.

Index Terms—Self-Balancing Robot, Model-Learning, Data Driven Control, Dissipativity-Based, Dynamic Conditions.

I. INTRODUCTION

Self-balancing two-wheeled robots provide high maneuverability, compact structure, and zero-radius turning capability, making them suitable for confined and dynamic environments. Their applications include warehouse automation, personal transportation, surveillance, logistics, and indoor navigation in spaces such as factories, mines, and crowded facilities.

Despite these advantages, the control of self-balancing robots remains challenging because the system is nonlinear, underactuated, and open-loop unstable. Practical operation is also affected by disturbances such as terrain variation, payload changes, actuator limitations, friction, and modeling uncertainties. Conventional PID and linear controllers can stabilize the robot near the upright equilibrium, but their performance may degrade when the system operates away from the nominal model or under external disturbances [1]. Therefore, robust and data-based control strategies are needed to improve balancing performance while reducing dependence on exact model knowledge.

This work develops a hybrid control framework for stabilizing a two-wheeled self-balancing robot. First, a baseline mathematical model is obtained from the linearized dynamics of the robot. The nonlinear dynamics are derived using Newton's second law and then linearized about the upright equilibrium to obtain a tractable state-space model [7]. This model is used to design a stabilizing controller through a

model-based linear matrix inequality (LMI) formulation. The model-based controller provides an initial stabilizing structure and establishes a reference for evaluating the subsequent data-driven designs.

Next, input-state data are generated from the controlled system and arranged into the data matrices. These matrices are then used to synthesize a controller directly from data. In the pure data-driven formulation, the closed-loop system is represented using measured trajectories, and the feedback gain is recovered without explicitly identifying the system matrices. This allows the control design to depend on collected system behavior rather than a fully known analytical model [4], [6].

To improve robustness, dissipativity is incorporated as an additional LMI-based performance constraint. In this formulation, dissipativity is not treated as an extra term in the data equation. Instead, the data equation provides the system representation, while the dissipativity condition imposes a bounded disturbance-to-performance-output gain [6]. The resulting LMI enforces stability while limiting the effect of disturbances on selected regulated outputs [8].

The proposed pipeline therefore progresses from model-based stabilization to data-driven control and finally to dissipativity-based robustness certification. This structure combines the reliability of model-based design, the flexibility of data-driven synthesis, and the safety guarantees of dissipativity theory. MATLAB simulations are conducted to compare the baseline data-driven controller and the dissipativity-enhanced controller under initial angle deviations, disturbance inputs, and parameter variations.

The main contributions of this work are summarized as follows:

- A linearized self-balancing robot model is derived to establish a model-based LMI stabilizing controller.
- Input-state trajectory data are collected and used to construct a data-driven stabilizing controller without explicit model identification.
- A dissipativity-based LMI condition is introduced to provide robustness against disturbance inputs.
- Data-assisted and direct data-driven dissipativity formulations are presented to connect measured data with robustness-based controller synthesis.

II. PRELIMINARY RESULTS

This section presents the theoretical control formulations used in the proposed workflow. Five propositions support the controller design process: model-based stabilization, direct data-driven stabilization, model-based dissipativity, data-assisted dissipativity, and direct data-driven dissipativity.

The model-based controller provides a stabilizing baseline and generates bounded input-state trajectories for data-driven synthesis. The collected matrices X_0 , X_1 , and U_0 are then used to compute a feedback gain without explicit model identification. Dissipativity-based constraints are further introduced to bound the disturbance-to-performance-output energy gain, either using matrices recovered from data or directly through measured data and disturbance bounds.

A. Model-Based State-Feedback Stabilization

Consider the linear time-invariant (LTI) system

$$\dot{\mathbf{x}}(t) = A\mathbf{x}(t) + Bu(t). \quad (1)$$

The objective is to design $u(t)$ such that the state $x(t)$ converges to equilibrium. A full-state feedback controller

$$u(t) = K\mathbf{x}(t), \quad (2)$$

where K is the controller gain matrix, leads to the closed-loop system

$$\dot{\mathbf{x}}(t) = (A + BK)\mathbf{x}(t). \quad (3)$$

Therefore, the stability of the closed-loop system depends on the eigenvalues of $A + BK$, and thus K needs to be selected strategically. For this purpose, the LMI problem formulated in the following proposition can be used.

Proposition 1: To stabilize the closed-loop system (3), the controller gain K has to be determined by the LMI problem

Find: Y, P

Sub. To: $P > 0, (AP + BY) + (AP + BY)^T < 0$.

where $K = YP^{-1}$.

B. Data-Driven LMI Control Design

Model-based LMI design requires knowledge of (A, B) , which may be unavailable due to uncertainties. To address this, stabilization is formulated directly using measured input-state data.

Consider the discrete-time LTI system

$$\mathbf{x}(t+1) = A\mathbf{x}(t) + Bu(t). \quad (4)$$

Collected trajectory data satisfy

$$X_1 = AX_0 + BU_0, \quad (5)$$

where

$$\begin{aligned} X_0 &= [x(0) \ x(1) \ \cdots \ x(T-1)], \\ X_1 &= [x(1) \ x(2) \ \cdots \ x(T)], \\ U_0 &= [u(0) \ u(1) \ \cdots \ u(T-1)]. \end{aligned} \quad (6)$$

For sufficiently rich data to design controllers, the rank condition

$$\text{rank} \begin{bmatrix} U_0 \\ X_0 \end{bmatrix} = n + m \quad (7)$$

is required. This condition is mild and will be easily satisfied with long enough (large T) data trajectories.

Proposition 2: To stabilize the closed-loop system (4) without relying on an explicit model, the state-feedback gain K can be computed directly from measured data by solving a data-dependent LMI problem

Find: Q

$$\text{Sub. To: } \begin{bmatrix} X_0Q & X_1Q \\ Q^T X_1^T & X_0Q \end{bmatrix} > 0,$$

where the feedback gain is recovered as

$$K = U_0Q(X_0Q)^{-1}. \quad (8)$$

C. Dissipativity-Based Robust Control Formulation

To account for external disturbances and uncertainties, the linearized system is augmented with a disturbance input $w(t) \in \mathbb{R}^p$ as

$$\dot{\mathbf{x}}(t) = A\mathbf{x}(t) + Bu(t) + Ew(t), \quad (9)$$

where E describes how the disturbance enters the system. Under the state-feedback control law (2) the disturbed closed-loop system becomes

$$\dot{\mathbf{x}}(t) = (A + BK)\mathbf{x}(t) + Ew(t). \quad (10)$$

A performance output is defined as

$$z(t) = C_z\mathbf{x}(t), \quad (11)$$

where C_z selects or weights the states whose response to disturbances should be attenuated.

To quantify disturbance attenuation, consider the quadratic storage function

$$V(x) = x^T P x, \quad P = P^T \succ 0. \quad (12)$$

The disturbance signal $w(t)$ is viewed as the external input, while the performance output $z(t)$ represents the regulated response whose energy should be limited. By selecting the quadratic supply rate

$$s(w, z) = \gamma^2 w^T w - z^T z, \quad (13)$$

the dissipativity condition corresponds to an $\mathcal{L}_2$ -gain requirement from $w(t)$ to $z(t)$. Therefore, if there exists a positive storage function such that

$$\dot{V}(x) \leq \gamma^2 w^T w - z^T z, \quad (14)$$

then the closed-loop system limits the amplification of disturbance energy at the performance output. Hence, reducing γ improves disturbance attenuation.

The following proposition gives a model-based LMI condition for dissipativity-based robust state-feedback design.

Proposition 3: Consider the disturbed continuous-time system (9) with performance output (11). If there exist matrices $X = X^T \succ 0$, Y , and a scalar $\gamma > 0$ such that

$$\begin{bmatrix} AX + BY + (AX + BY)^T & E & (C_z X)^T \\ E^T & -\gamma^2 I & 0 \\ C_z X & 0 & -I \end{bmatrix} \prec 0,$$

then the state-feedback gain

$$K = YX^{-1} \quad (15)$$

guarantees closed-loop dissipativity with respect to the supply rate (13).

D. Data-Assisted Dissipativity-Based Control Formulation

The model-based dissipativity controller requires accurate knowledge of the system matrices A and B . Since these matrices may be affected by modeling uncertainty, a data-assisted formulation is introduced to reduce direct dependence on the analytical model. In this approach, measured input-state data are first used to recover an estimated model, and the dissipativity-based LMI is then applied to the recovered matrices. This keeps the controller synthesis numerically practical while allowing the system representation to be obtained from data.

For the data-assisted dissipativity design, the continuous-time linearized model is first discretized as

$$x(k+1) = Ax(k) + Bu(k), \quad (16)$$

where $x(k) \in \mathbb{R}^n$ is the state vector and $u(k) \in \mathbb{R}^m$ is the control input.

In practical implementation, the system matrices A and B are not assumed to be directly available. Instead, they are recovered from measured input-state data.

The performance output is defined as

$$z(k) = C_z x(k), \quad (17)$$

where C_z is the performance-output matrix. A state-feedback controller of the form

$$u(k) = Kx(k) \quad (18)$$

is considered.

Under the rank condition (7), an estimated discrete-time representation of the system can be obtained as

$$\begin{bmatrix} \hat{B} & \hat{A} \end{bmatrix} = X_1 \begin{bmatrix} U_0 \\ X_0 \end{bmatrix}^\dagger. \quad (19)$$

The recovered model is then used for dissipativity-based controller synthesis. Including the disturbance channel, the estimated disturbed closed-loop system is written as

$$x(k+1) = (\hat{A} + \hat{B}K)x(k) + Ew(k), \quad (20)$$

where $w(k)$ is the external disturbance and E defines how the disturbance enters the system.

The objective is to design K such that the closed-loop system is stable and the disturbance effect from $w(k)$ to $z(k)$ is attenuated. This is imposed using the supply rate

$$s(w, z) = \gamma^2 w(k)^T w(k) - z(k)^T z(k), \quad (21)$$

which corresponds to an $\mathcal{L}_2$ -gain bound from the disturbance input to the performance output.

The following proposition gives the data-assisted dissipativity condition used for controller synthesis.

Proposition 4: Consider the discrete-time system (16), where the system matrices A and B are not assumed to be known explicitly. From the measured data matrices X_0 , X_1 , and U_0 (6), assume that the rank condition (7) is satisfied.

If there exist matrices $X = X^T \succ 0$, Y , and a scalar $\gamma > 0$ such that

$$\begin{bmatrix} X & \hat{A}X + \hat{B}Y & E & 0 \\ (\hat{A}X + \hat{B}Y)^T & X & 0 & XC_z^T \\ E^T & 0 & \gamma^2 I & 0 \\ 0 & C_z X & 0 & I \end{bmatrix} \succ 0,$$

then the feedback gain (15) provides a dissipativity-based robust controller for the estimated closed-loop system. The resulting implementation uses the control law (18).

E. Data-Driven Dissipativity-Based Control Formulation

The data-assisted method still depends on a recovered model $\hat{A}$ and $\hat{B}$. To eliminate this intermediate identification step, the dissipativity condition can instead be formulated directly using the measured data matrices. This provides a fully data-driven robustness formulation, where the controller is synthesized from trajectory data while still enforcing a disturbance-to-performance gain bound.

This formulation is especially relevant when the collected data are affected by disturbances. In that case, the data equation no longer satisfies the nominal relation $X_1 = AX_0 + BU_0$, but instead contains an additional disturbance term. Therefore, the dissipativity condition must be imposed while accounting for the uncertainty introduced by the unknown disturbance sequence.

For the data-driven design, the continuous-time linearized model is discretized as

$$x(k+1) = Ax(k) + Bu(k) + w(k), \quad (22)$$

where $x(k) \in \mathbb{R}^n$ is the state vector, $u(k) \in \mathbb{R}^m$ is the control input, and $w(k) \in \mathbb{R}^p$ is the external disturbance. The performance output is defined as (17), where C_z is the performance-output matrix.

Substituting (18) into (22) gives the disturbed closed-loop system

$$x(k+1) = (A + BK)x(k) + w(k). \quad (23)$$

Instead of relying explicitly on the system matrices A and B , the data-driven formulation uses measured input and state trajectories. If the collected data are affected by disturbance, then the data satisfy

$$X_1 = AX_0 + BU_0 + W_0, \quad (24)$$

where X_0 , X_1 , and U_0 (6) and

$$W_0 = [w(0) \ w(1) \ \cdots \ w(T-1)] \quad (25)$$

is the disturbance sequence during data collection. Since W_0 is generally unknown, it is assumed to be bounded as

$$W_0 W_0^T \preceq \delta_w I, \quad (26)$$

where $\delta_w > 0$ is the disturbance-energy bound.

The objective is to design the feedback gain K directly from the measured data such that the closed-loop system is stable and the effect of $w(k)$ on $z(k)$ is attenuated. This is enforced using the supply rate (21), which corresponds to an $\mathcal{L}_2$ -gain bound from disturbance input to performance output.

Proposition 5: To establish a fully data-driven dissipativity-based control formulation. The disturbance sequence during data collection is denoted by (25). Define

$$\Phi_w = \begin{bmatrix} I & X_1 \\ 0 & -X_0 \\ 0 & -U_0 \end{bmatrix} \begin{bmatrix} \delta_w I & 0 \\ 0 & -I \end{bmatrix} \begin{bmatrix} I & X_1 \\ 0 & -X_0 \\ 0 & -U_0 \end{bmatrix}^T. \quad (27)$$

If there exist matrices $\bar{P} = \bar{P}^T \succ 0$, $\bar{K}$, and a scalar $\lambda > 0$

such that

$$\begin{bmatrix} \Psi & \Lambda \\ \Lambda^T & \bar{Q}_p \end{bmatrix} \succeq 0, \quad (28)$$

where

$$\begin{aligned} \Psi &= \begin{bmatrix} I & C_z \bar{P} & 0 \\ (C_z \bar{P})^T & \bar{P} & 0 \\ 0 & 0 & \gamma^2 I \end{bmatrix}, \\ \Lambda &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & \bar{K}^T \\ I & 0 & 0 \end{bmatrix}, \\ \bar{Q}_p &= \begin{bmatrix} \bar{P} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} - \lambda \Phi_w. \end{aligned} \quad (29)$$

then the controller

$$K = \bar{K} \bar{P}^{-1} \quad (30)$$

renders the closed-loop system dissipative with respect to the supply rate (i.e., the disturbance-to-performance-output gain is bounded by γ).

III. PROBLEM DESCRIPTION

Two-wheeled self-balancing robots are nonlinear, underactuated, and inherently unstable systems, modeled as wheeled inverted pendulums as stated in [1] [7]. The dynamics are given by

$$(M + m)\ddot{x} + b\dot{x} + ml\ddot{\theta} \cos \theta - ml\dot{\theta}^2 \sin \theta = F \quad (31)$$

$$(I + ml^2)\ddot{\theta} + mgl \sin \theta = -ml\ddot{x} \cos \theta \quad (32)$$

where M is cart mass, m body mass, l pendulum length, x position, θ tilt, F wheel force. Linearization around $\theta = 0$ yields unstable poles, which are sensitive to parameters [1] [7].

To facilitate control design, the nonlinear dynamics in (31) and (32) are embedded within a structured control architecture as shown in Fig. 1 [1]. These equations define the nonlinear plant, where the wheel-generated force F acts as the input and the states $(x, \dot{x}, \theta, \dot{\theta})$ evolve according to coupled cart-pendulum dynamics.

In Fig. 1, the plant is shown as a subsystem, while the remaining blocks represent sensing, control, and actuation. IMU measurements of θ and $\dot{\theta}$ are fed to the controller, which computes the control input. This input is then distributed to the wheels via allocation and motor control, closing the feedback loop.

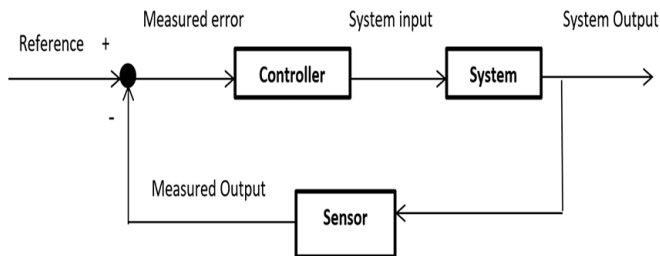


Fig. 1. Control System Block Diagram

This representation abstracts the nonlinear dynamics while preserving their influence on the closed-loop behavior.

Key challenges include:

- **Instability:** Fast unstable eigenvalues demand high-bandwidth control.
- **Disturbances:** Pushes/terrain cause drift; linear controllers fail beyond small angles.
- **Uncertainties:** Friction, payload shifts alter CoM/inertia; neglected couplings degrade performance.
- **Nonlinearities:** Large tilts require global control; MPC computationally heavy.
- **Drift/Positioning:** Balance induces velocity bias without secondary loops.

A. Model-Based Limitations

Model-based control relies on accurate system models, which are often difficult to obtain for complex systems due to nonlinearities, uncertainties, and time-varying dynamics. Consequently, modeling errors can degrade performance and compromise stability [2] [3]. Most approaches rely on linearization around equilibrium points, limiting their validity to small deviations and leading to degraded performance under large disturbances or varying conditions [2] [1]. Although robust methods attempt to address uncertainties, they introduce conservatism due to difficulty in accurately bounding them, resulting in suboptimal performance [5].

B. Data-Driven Control Limitations

Data-driven control methods have several limitations. They require sufficiently informative data, typically ensured through persistent excitation; inadequate excitation can lead to infeasible designs [6]. In practice, excitation is constrained by actuator limits and safety requirements, making performance highly dependent on data quality, especially in unstable systems [2] [3]. These methods are also sensitive to measurement noise, as controller synthesis relies directly on data, while robust formulations depend on assumptions about noise bounds that are difficult to estimate [6] [1]. Moreover, many approaches rely on linear assumptions, reducing effectiveness under large disturbances or operation outside the data region [2] [1]. Finally, data-driven control often involves large-scale optimization and parameter tuning, increasing computational burden and limiting scalability and real-time applicability [5].

C. Hybrid Solution Motivation

The proposed framework combines model-based, data-assisted, and data-driven control design to improve the reliability of the self-balancing robot under dynamic conditions. The motivation for this hybrid structure is summarized as follows:

- **Model-Based LMI Control:** Provides a stable baseline controller using the linearized mathematical model.
- **Data-Driven LMI Control:** Uses the generated input-output data matrices directly for controller synthesis without explicitly relying on the system model.

- **Dissipativity-Based Robust Control:** Introduces disturbance attenuation and performance-output constraints to improve robustness.
- **Data-Assisted Dissipativity Control:** Recovers $\hat{A}$ and $\hat{B}$ from collected data and uses them in a dissipativity-based LMI.
- **Data-Driven Dissipativity Control:** Incorporates dissipativity constraints directly into the data-driven formulation using the collected data matrices, disturbance channel, and performance output.

Therefore, when we look into the flowchart, Fig. 2.

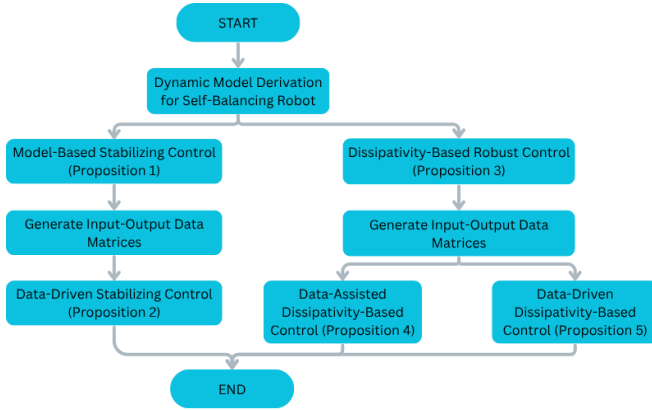


Fig. 2. Project Flowchart

IV. DYNAMIC MODEL DERIVATION FOR SELF-BALANCING ROBOT

The mathematical model of the two-wheeled self-balancing robot is based on the wheeled inverted pendulum configuration, with Fig. 3 as the reference. The system dynamics are derived using Newton's Second Law, $\sum F = ma$ and $\sum M = I\alpha$. The control objective is to maintain the upright equilibrium, where all accelerations are zero, i.e., $\sum F = 0$ and $\sum M = 0$.

Unlike the classical cart-pendulum system, the robot employs two independently actuated motors in a differential-drive configuration. This enables both forward motion and

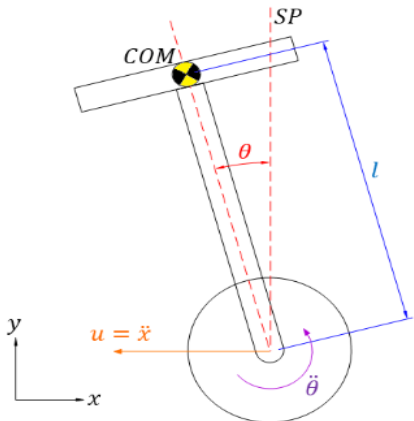


Fig. 3. Dynamics of Self-Balancing Robot

yaw control, introducing additional coupling between translational, rotational, and steering dynamics.

Applying Newton's laws to the system in Fig. 3, the force and moment balances are obtained. The horizontal force balance is given by

$$\sum f_x = 0 \iff F - b\dot{x} - (M + m)\ddot{x} - ml\ddot{\theta} \cos \theta + ml\dot{\theta}^2 \sin \theta = 0 \quad (33)$$

which captures the driving force, viscous friction, inertial effects, and coupling between translational and rotational motion.

The vertical force balance is expressed as

$$\sum f_y = 0 \iff Mg + mg - ml\ddot{\theta} \sin \theta - ml\dot{\theta}^2 \cos \theta = 0 \quad (34)$$

which enforces the no-lift-off constraint through gravitational and inertial contributions.

The pitching dynamics about the wheel axle are obtained from the moment balance

$$\sum M_A = 0 \iff mgl \sin \theta + I\ddot{\theta} + ml\ddot{x} \cos \theta = 0 \quad (35)$$

where the gravitational torque, rotational inertia, and coupling with chassis acceleration determine the balancing behavior.

Additionally, the yaw dynamics arise from differential wheel forces and are given by

$$\sum M_z = 0 \iff l_w(F_R - F_L) - I_z\ddot{\psi} = 0 \quad (36)$$

which enables steering through the difference in left and right wheel forces.

Together, (33)–(36) describe the full nonlinear dynamics of the system. In this work, emphasis is placed on the longitudinal and pitching dynamics (33) and (35), as they directly govern balance, while yaw dynamics are acknowledged but not explicitly considered to reduce model complexity.

A. Nonlinear State-Space Model

The nonlinear dynamics of the system are expressed in a control-affine state-space form as

$$\dot{\mathbf{x}} = f(\mathbf{x}) + g(\mathbf{x})u. \quad (37)$$

Here, u represents the equivalent longitudinal driving force applied through the wheel actuators. In hardware, this input is generated by the left and right motor torques rather than applied directly to the chassis. If τ_L and τ_R are the wheel torques and r_w is the wheel radius, then

$$u = F = \frac{\tau_L + \tau_R}{r_w}.$$

For straight-line balancing, the torque can be distributed equally between the two wheels as

$$\tau_L = \tau_R = \frac{r_w u}{2}.$$

Thus, during execution, the controller computes u , and the robot program maps this command into motor torque or PWM signals. Since this work focuses on longitudinal

balance control, yaw motion and differential wheel allocation are not explicitly considered.

The state vector is defined as

$$\mathbf{x} = \begin{bmatrix} x \\ \dot{x} \\ \theta \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}. \quad (38)$$

From this definition, two of the state equations follow directly:

$$\dot{x}_1 = x_2, \quad \dot{x}_2 = \ddot{x}, \quad \dot{x}_3 = x_4, \quad \dot{x}_4 = \ddot{\theta}.$$

Rearranging the (33) and (35), the coupled dynamics can be written as

$$\begin{aligned} (M+m)\ddot{x} - ml\ddot{\theta}\cos\theta &= r_1(x, u), \\ (I+ml^2)\ddot{\theta} - ml\ddot{x}\cos\theta &= r_2(x), \end{aligned}$$

where

$$r_1(x, u) = u - bx_2 - mlx_4^2 \sin x_3, \quad r_2(x) = -mgl \sin x_3.$$

These equations represent a coupled second-order system, where translational and rotational accelerations are interdependent. To explicitly solve for $\ddot{x}$ and $\ddot{\theta}$, the system is written in matrix form as

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} \ddot{x} \\ \ddot{\theta} \end{bmatrix} = \begin{bmatrix} r_1 \\ r_2 \end{bmatrix}, \quad (39)$$

with

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \begin{bmatrix} M+m & -ml\cos\theta \\ ml\cos\theta & I+ml^2 \end{bmatrix}.$$

By inverting this matrix, the accelerations are obtained as

$$\ddot{x} = \frac{a_{11}r_1 + a_{12}r_2}{\Delta}, \quad \ddot{\theta} = \frac{a_{21}r_1 + a_{22}r_2}{\Delta},$$

where $\Delta = a_{11}a_{22} - a_{12}a_{21}$. These expressions explicitly show how the nonlinear forces, gravity, and control input influence the system accelerations.

Substituting these results into the state equations yields the nonlinear state-space model

$$\dot{\mathbf{x}} = \underbrace{\begin{bmatrix} x_2 \\ f_1(x) \\ x_4 \\ f_2(x) \end{bmatrix}}_{f(x)} + \underbrace{\begin{bmatrix} 0 \\ g_1(x) \\ 0 \\ g_2(x) \end{bmatrix}}_{g(x)} u, \quad (40)$$

where $f(x)$ represents the natural nonlinear dynamics, including gravitational and coupling effects, and $g(x)$ describes how the input force influences the system through the mass-inertia coupling.

This control-affine formulation separates the intrinsic system behavior from the control input, making it suitable for nonlinear control design. With the nonlinear state-space model established, the next step is to linearize the system about the upright equilibrium to obtain a tractable model for controller synthesis.

B. Linearization of Nonlinear Equations

To enable linear control design, the nonlinear model (40) is linearized about the upright equilibrium to obtain

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u. \quad (41)$$

The angular coordinate is redefined as $\theta = \pi + \phi$, where $\phi \approx 0$. Using small-angle approximations,

$$\sin(\pi + \phi) \approx -\phi, \quad \cos(\pi + \phi) \approx -1,$$

the nonlinear dynamics are simplified near the equilibrium. Substituting and neglecting higher-order terms yields

$$(M+m)\ddot{x} + ml\ddot{\phi} + b\dot{x} = u, \quad (I+ml^2)\ddot{\phi} - mgl\phi = ml\ddot{x}.$$

Define $a = M+m$, $c = ml$, $d = I+ml^2$, to rewrite

$$a\ddot{x} + c\ddot{\phi} = u - b\dot{x}, \quad -c\ddot{x} + d\ddot{\phi} = mgl\phi.$$

Solving gives

$$\ddot{x} = \frac{d(u - b\dot{x}) - cmgl\phi}{\Delta}, \quad \ddot{\phi} = \frac{c(u - b\dot{x}) + amgl\phi}{\Delta},$$

where $\Delta = ad + c^2$. With state vector (38) the linearized state-space model is

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -\frac{db}{\Delta} & -\frac{cmgl}{\Delta} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & -\frac{cb}{\Delta} & \frac{amgl}{\Delta} & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ \frac{d}{\Delta} \\ 0 \\ \frac{c}{\Delta} \end{bmatrix}. \quad (42)$$

The matrix $\mathbf{A}$ captures the coupled dynamics and gravitational instability, while $\mathbf{B}$ describes the input influence. This model forms the basis for subsequent state-feedback and LMI-based control design.

V. LMI-BASED CONTROL DESIGN

A. Model-Based to Data-Driven Control Flow

In Fig.4 control path begins with the baseline mathematical model of the self-balancing robot. The nonlinear dynamics are linearized around the upright equilibrium point to obtain the discrete-time state-space representation. This model provides the system matrices $\mathbf{A}$ and $\mathbf{B}$, which are used for the initial model-based controller design.

A model-based LMI controller is then synthesized using the state-space model. Solving the LMI provides the stabilizing feedback gain $\mathbf{K}$, which acts as the baseline controller. The closed-loop system is then simulated using $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{K}$ to generate input-output data.

The collected trajectories are arranged into the data matrices $\mathbf{X}_0$, $\mathbf{X}_1$, and $\mathbf{U}_0$ (6), where $\mathbf{X}_0$ contains the current state samples, $\mathbf{X}_1$ contains the next-state samples, and $\mathbf{U}_0$ contains the corresponding input samples. These matrices are then used in the data-driven LMI formulation to synthesize a controller directly from data, without explicitly using the model matrices in the final design step.

B. Dissipativity-Based Data-Driven Control Flow

In Fig.5 control path extends the baseline model-based design by introducing a dissipativity-based robust controller. Starting from the baseline mathematical model, the system is first represented in discrete-time state-space form. A dissipativity-based LMI is then formulated to include disturbance attenuation and performance-output constraints.

Solving this LMI provides the system matrices $\mathbf{A}$, $\mathbf{B}$, and the dissipativity-based feedback gain $\mathbf{K}$. This controller is then used to simulate the closed-loop system and generate input-output trajectory data. From the collected data, the matrices $\mathbf{X}_0$, $\mathbf{X}_1$, and $\mathbf{U}_0$ (6) are constructed, where $\mathbf{X}_0$

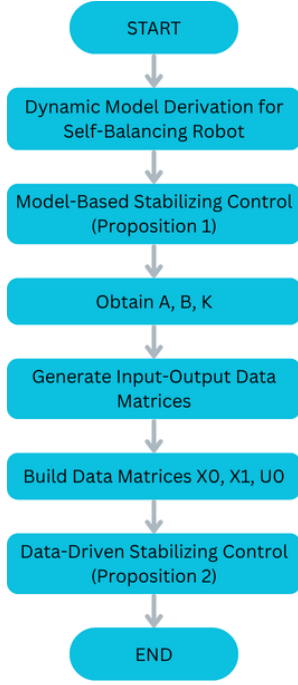


Fig. 4. Model-based to data-driven stabilizing control flow.

contains the current state samples, X_1 contains the next-state samples, and U_0 contains the applied control inputs.

After the data matrices are generated, two dissipativity-related data-driven paths are considered. In the data-assisted branch, the matrices $\hat{A}$ and $\hat{B}$ are recovered from the collected data and used in a dissipativity-based LMI. In the direct data-driven dissipativity branch, the disturbance input W_0 is included along with the collected data matrices to formulate the dissipativity-based controller directly from data.

C. Input-State Data Matrix

The data-driven design uses the measured input-state matrices X_0 , X_1 , and U_0 (6). Each column represents one sampled transition of the system. The rank condition in (7) is checked to ensure that the collected trajectories are sufficiently informative for controller synthesis.

For safe data collection, the system is excited using a temporary stabilizing controller with an added excitation signal,

$$u(k) = K_{\text{temp}}x(k) + u_{\text{exc}}(k), \quad (43)$$

where K_{temp} keeps the trajectories bounded and $u_{\text{exc}}(k)$ provides the required excitation for data-driven control design.

D. Data-Assisted Dissipativity-Based Controller

Dissipativity is imposed through the inequality (14), where $V(x) = x^T Px$ and performance output (17). Proposition 4 and Proposition 5 enforce the same disturbance-to-performance attenuation objective, but differ in how the dynamics are represented.

In Proposition 4, the data matrices are used to recover (19). The closed-loop system is then written as (20). With the substitution $Y = KX$, the closed-loop product becomes

$$(\hat{A} + \hat{B}K)X = \hat{A}X + \hat{B}Y, \quad (44)$$

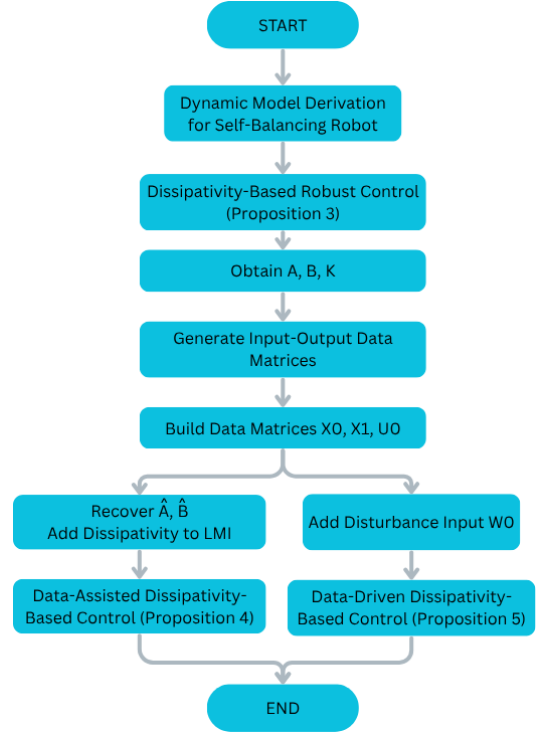


Fig. 5. Dissipativity-based data-assisted and data-driven control flow.

which gives a standard convex bounded-real LMI. This makes the data-assisted formulation easier to solve numerically using YALMIP and SDPT3.

In Proposition 5, the model-recovery step is avoided. However, disturbed data satisfy (24), where W_0 is unknown. Hence, the LMI must hold for all (A, B) consistent with the data and the disturbance bound (26). This introduces additional dependence on the data-consistency matrix, the bound δ_w , and an S-lemma multiplier λ . As a result, the direct data-driven dissipativity formulation is more sensitive to data conditioning, scaling, and the selected disturbance bound.

Therefore, Proposition 4 is chosen for the simulation implementation because it provides a practical and numerically stable dissipativity design while still using data to construct the system representation. Proposition 5 remains the fully data-driven theoretical extension, but it is more demanding to implement reliably for the available data.

VI. SIMULATION

To evaluate controller performance, MATLAB simulations are conducted using the linearized state-space model in (42). The simulations are initialized with

$$\mathbf{x}(0) = [0 \quad 0 \quad 45^\circ \quad 0]^T,$$

where the cart position, cart velocity, and angular velocity are zero, while the body angle starts at 45° . This initial condition tests the controller's ability to recover from a large tilt and return the robot to the upright equilibrium.

1) *Baseline Simulation and Data Generation:* The baseline simulation implements the model-based stabilizing controller from Proposition 1. The continuous-time linearized model is discretized using sampling time $T_s = 0.01$ s, and a

temporary stabilizing feedback controller is applied to keep the trajectories bounded during data collection.

To generate data suitable for Proposition 2, a persistently exciting input is added to the temporary controller. The resulting trajectories are stored as the data matrices X_0 , X_1 , and U_0 in (6). The rank condition in (7) is then checked to confirm that the collected data are sufficiently informative for data-driven controller synthesis.

2) *Pure Data-Driven Controller Design*: The pure data-driven controller implements Proposition 2 using only the saved matrices X_0 , X_1 , and U_0 . To reduce redundancy in the data set, linearly independent columns are selected from

$$Z = \begin{bmatrix} U_0 \\ X_0 \end{bmatrix}, \quad (45)$$

forming the compressed matrices X_{0c} , X_{1c} , and U_{0c} while preserving the required rank condition.

The data-driven LMI in Proposition 2 is then solved using YALMIP and SDPT3. The gain is recovered directly from the compressed data, and the closed-loop stability is verified by checking that the eigenvalues of the data-based closed-loop matrix lie inside the unit circle. The state responses, control input, and robot animation are generated to visualize recovery from the initial 45° body-angle deviation.

3) *Model-Based Dissipativity Controller*: The model-based dissipativity controller implements Proposition 3. In this case, the discretized model matrices are used directly in the dissipativity-based bounded-real LMI. The disturbance channel and performance output are introduced to evaluate robustness against external disturbances.

The performance output is first selected as

$$C_z = \text{diag}(0, 0, 1, 0), \quad (46)$$

which emphasizes only the body angle. This stabilizes the angular response, but the robot position can drift significantly. Therefore, the final performance weighting is selected as

$$C_z = \text{diag}(0.2, 0.05, 1, 0.2). \quad (47)$$

This modified weighting includes position, velocity, body angle, and angular velocity, improving both balancing and translational recovery.

After solving the LMI in Proposition 3, the controller is tested from the same initial 45° body-angle deviation. Actuator saturation is included as

$$u_{\text{sat}}(k) = \text{sat}(K_{\text{diss}}x(k), u_{\text{max}}), \quad (48)$$

where $u_{\text{max}} = 20$, to obtain a more realistic simulation response.

4) *Data-Assisted Dissipativity Controller*: The data-assisted dissipativity controller corresponds to Proposition 4. In this implementation, trajectories generated from the model-based dissipativity controller are used to form the data matrices X_0 , X_1 , and U_0 . These data are then used to recover an estimated model, after which the dissipativity-based LMI is applied to the recovered matrices as described in Proposition 4.

For the final data-assisted stabilization step, the standard data-driven LMI is solved using the compressed data matri-

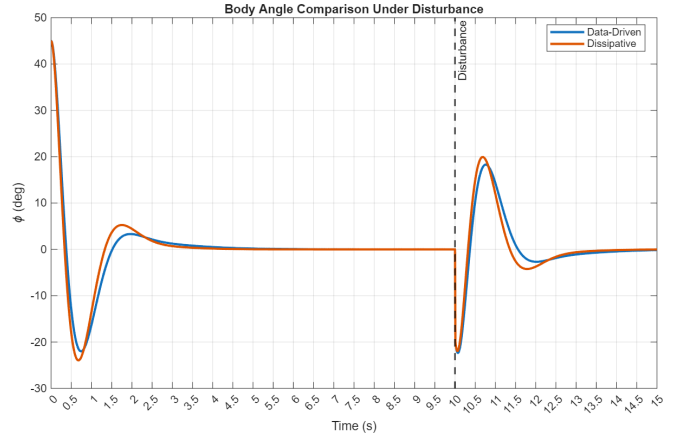


Fig. 6. Body angle comparison under disturbance.

ces. A decision matrix Q is introduced, with

$$P_{\text{data}} = X_{0c}Q. \quad (49)$$

To improve the stability margin, the LMI is modified as

$$\begin{bmatrix} \alpha P_{\text{data}} & X_{1c}Q \\ Q^T X_{1c}^T & \alpha P_{\text{data}} \end{bmatrix} \succ 0, \quad (50)$$

where $\alpha = 0.98$. Since $\alpha < 1$, the closed-loop poles are encouraged to lie further inside the unit circle while avoiding excessive conservatism.

5) *Comparison Under Disturbance*: To compare disturbance rejection performance, the controllers obtained from Proposition 2 and the dissipativity-enhanced implementation based on Propositions 4 are tested under the same initial condition and disturbance input. The initial state is

$$x(0) = [0 \ 0 \ 45^\circ \ 0]^T. \quad (51)$$

An impulse-like disturbance is applied at $t = 10$ s by perturbing the body angle and angular velocity states:

$$\Delta x = [0 \ 0 \ -20^\circ \ -80^\circ/\text{s}]^T. \quad (52)$$

The control signals are saturated to represent actuator limitations:

$$u_{\text{sat}}(k) = \text{sat}(u(k), 20). \quad (53)$$

The comparison is performed using maximum body-angle deviation, angular velocity, control input, final body angle, final state norm, and post-disturbance dissipativity energy. These metrics show how the standard data-driven controller and the dissipativity-enhanced controller recover under identical disturbance conditions.

VII. RESULTS AND DISCUSS

The standard data-driven and dissipativity-based controllers are compared under an external disturbance applied at $t = 10$ s. Both controllers start from the same 45° body-angle deviation and successfully stabilize the robot before and after the disturbance.

From Figs. 6–9, both controllers recover after the disturbance. The standard data-driven controller gives a slightly smaller immediate body-angle peak, while the dissipativity-based controller applies a more aggressive correction. This results in a slightly larger angular-velocity transient, but also faster recovery in the position and velocity responses.

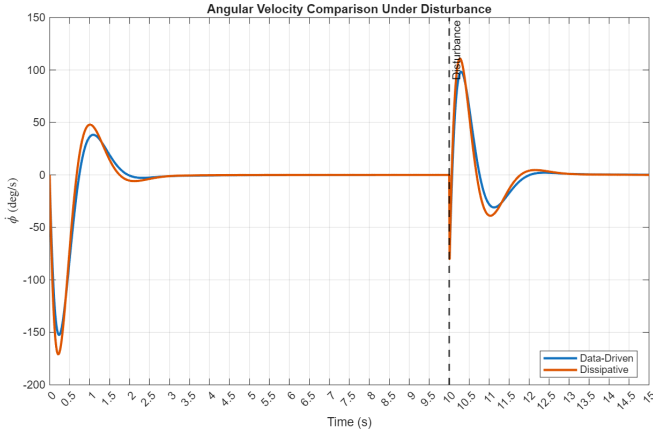


Fig. 7. Angular velocity comparison under disturbance.

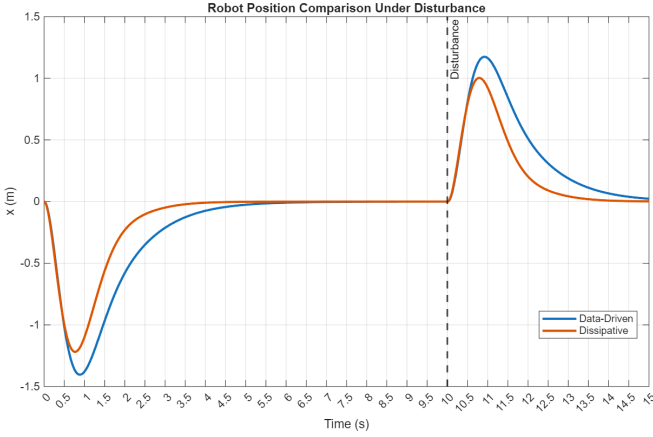


Fig. 8. Robot position comparison under disturbance.

In particular, the dissipativity-based controller limits the post-disturbance cart displacement more effectively than the standard data-driven controller.

The control-input response in Fig. 10 shows that the dissipativity-based controller uses a higher maximum control effort of 16.2469, compared with 14.2869 for the standard data-driven controller. This stronger control action improves disturbance attenuation, as shown in Fig. 11. The dissipativity-based controller reduces the performance output norm $\|z(k)\|$ faster after the disturbance.

The cumulative post-disturbance energy in Fig. 13 further confirms this behavior. The standard data-driven controller gives a post-disturbance energy of 19.8043, while the dissipativity-based controller reduces it to 17.9942. Therefore, although the dissipativity-based controller does not minimize every peak state response, it provides better overall energy-based disturbance rejection.

VIII. CONCLUSION

This work presented a model-based and data-driven control framework for a two-wheeled self-balancing robot under dynamic conditions. The nonlinear robot dynamics were modeled as a wheeled inverted pendulum and linearized around the upright equilibrium to obtain a state-space model for LMI-based controller synthesis.

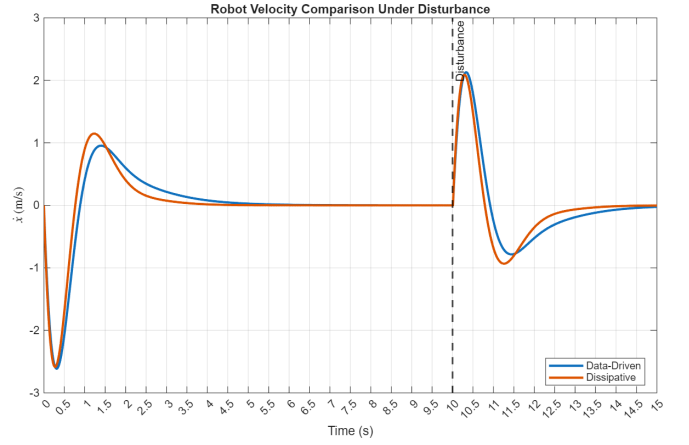


Fig. 9. Robot velocity comparison under disturbance.

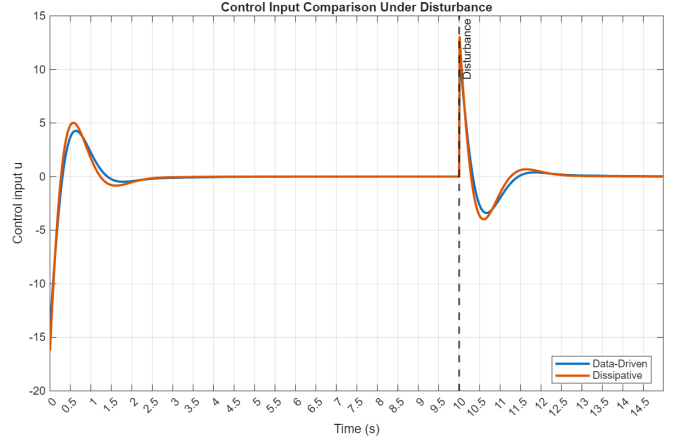


Fig. 10. Control input comparison under disturbance.

A pure data-driven controller was then designed using the measured data matrices X_0 , X_1 , and U_0 (6), without explicitly using the system matrices in the final synthesis step. The rank condition verified that the collected trajectories were sufficiently informative, and the controller successfully stabilized the robot from an initial 45° body-angle deviation.

To improve robustness, dissipativity-based control was introduced using the performance output $z(k) = C_z x(k)$. The choice of C_z was important, since weighting only the body angle caused larger position drift, while including position, velocity, body angle, and angular velocity improved both balancing and translational recovery. The data-assisted approach then used trajectories generated from the dissipativity-based controller to synthesize a final data-driven controller, providing a practical bridge between model-based robustness and data-driven implementation.

Simulation results showed that both controllers stabilized the robot before and after the disturbance applied at $t = 10$ s. The standard data-driven controller produced a slightly smaller immediate body-angle peak, but the dissipativity-based controller achieved lower post-disturbance energy, reducing it from 19.8043 to 17.9942. This confirms improved robustness in the dissipativity sense, at the cost of slightly higher control effort.

Overall, the proposed framework shows that combining

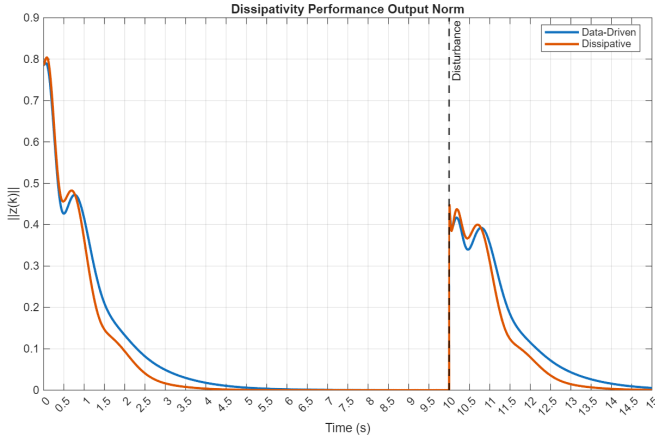


Fig. 11. Dissipativity performance output norm under disturbance.

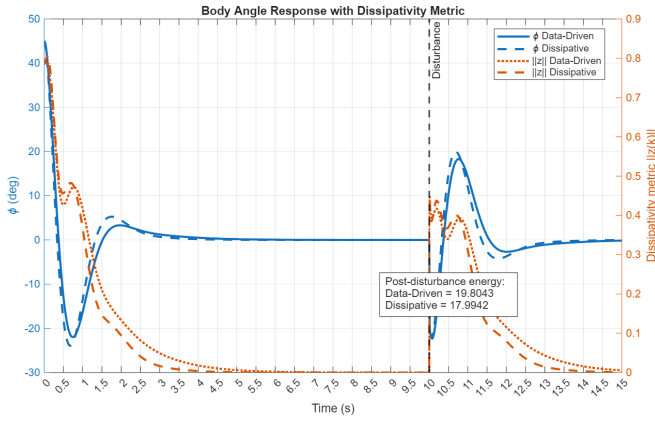


Fig. 12. Body angle response with dissipativity metric.

model-based control, data-driven synthesis, and dissipativity-based constraints can improve disturbance rejection for self-balancing robots. Future work will focus on hardware implementation, nonlinear data-driven control, output-feedback design, and real-time disturbance estimation.

All MATLAB source codes, simulation results, generated plots, and robot animation files are available in the project GitHub repository.¹

APPENDIX

A. Proof of Proposition 1

Proof: Using Lyapunov theory, consider $V(x) = x^T P x$, with $P \succ 0$. Asymptotic stability is guaranteed if

$$(A + BK)^T P + P(A + BK) < 0. \quad (54)$$

This ensures energy decay and convergence to equilibrium. However, the condition is nonlinear in K and P , motivating its reformulation as an LMI for convex optimization.

The Lyapunov stability condition ensures asymptotic stability of the closed-loop system. However, it is bilinear in P and K , making it non-convex.

To obtain a convex formulation, define

$$Y = KP \iff K = YP^{-1} \quad (55)$$

¹<https://github.com/your-username/your-repository>

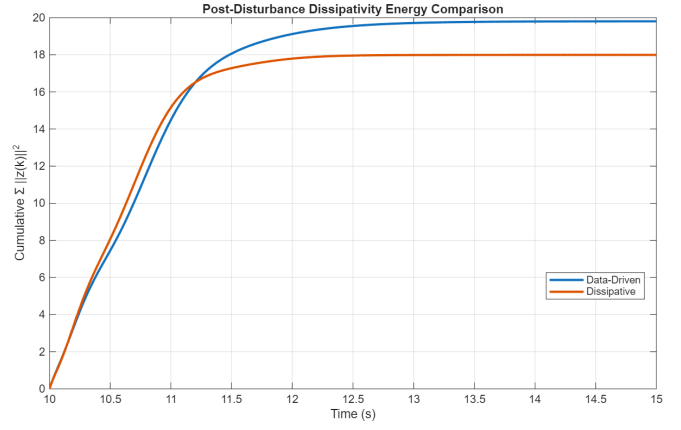


Fig. 13. Post-disturbance cumulative dissipativity energy comparison.

Substituting (55) into (54) and multiplying by P gives

$$(AP + BY) + (AP + BY)^T < 0, \quad P \succ 0. \quad (56)$$

This condition is linear in P and Y , forming an LMI. The LMI can be solved via semidefinite programming, and the feedback gain is recovered as (55). ■

B. Proof of Proposition 2

Proof: To ensure asymptotic stability, consider a Lyapunov function $V(x) = x^T P x$ for (4), with $P \succ 0$. Since P is positive definite, the Lyapunov function satisfies

$$V(x(k)) > 0, \quad \forall x(k) \neq 0, \quad (57)$$

and $V(0) = 0$. For the closed-loop system

$$x(k+1) = Rx(k), \quad (58)$$

where

$$R = A + BK, \quad (59)$$

the change in the Lyapunov function over one sampling step is

$$\begin{aligned} \Delta V &= V(x(k+1)) - V(x(k)) \\ &= x(k+1)^T P x(k+1) - x(k)^T P x(k). \end{aligned}$$

Substituting (58) gives

$$\begin{aligned} \Delta V &= (Rx(k))^T P (Rx(k)) - x(k)^T P x(k) \\ &= x(k)^T (R^T P R - P) x(k). \end{aligned} \quad (60)$$

Equivalently,

$$P - Q^T X_1^T P^{-1} X_1 Q \succ 0. \quad (61)$$

Applying the Schur complement, the above inequality is equivalent to

$$\begin{bmatrix} P & X_1 Q \\ Q^T X_1^T & P \end{bmatrix} \succ 0.$$

Finally, replacing P with $X_0 Q$ yields the data-dependent LMI

$$\begin{bmatrix} X_0 Q & X_1 Q \\ Q^T X_1^T & X_0 Q \end{bmatrix} \succ 0. \quad (62)$$

This condition is linear in the decision variable Q and depends only on measured data. Solving this LMI provides Q , from which the stabilizing feedback gain is recovered as (8).

Thus, the controller is obtained directly from data without identifying the system matrices, while still guaranteeing

closed-loop stability. ■

C. Proof of Proposition 3

Proof: To enforce robustness, the closed-loop system is required to satisfy the $\mathcal{L}_2$ -gain dissipativity condition from $w(t)$ to $z(t)$. Consider the quadratic storage function (12). Let's use (59). Then the disturbed closed-loop dynamics can be written as

$$\dot{x}(t) = Rx(t) + Ew(t). \quad (63)$$

The time derivative of the storage function is

$$\dot{V}(x) = \dot{x}^T Px + x^T P \dot{x}. \quad (64)$$

Substituting the closed-loop dynamics gives

$$\dot{V}(x) = x^T (R^T P + PR)x + 2x^T PEw.$$

Since performance output is (11), we have

$$z^T z = x^T C_z^T C_z x. \quad (65)$$

Substituting these expressions into (14) yields

$$x^T (R^T P + PR)x + 2x^T PEw \leq \gamma^2 w^T w - x^T C_z^T C_z x.$$

Rearranging all terms to the left-hand side gives

$$x^T (R^T P + PR + C_z^T C_z)x + 2x^T PEw - \gamma^2 w^T w \leq 0.$$

This inequality can be written in quadratic form as

$$\begin{bmatrix} x \\ w \end{bmatrix}^T \begin{bmatrix} R^T P + PR + C_z^T C_z & PE \\ E^T P & -\gamma^2 I \end{bmatrix} \begin{bmatrix} x \\ w \end{bmatrix} \leq 0.$$

Therefore, a sufficient condition for dissipativity is

$$\begin{bmatrix} R^T P + PR + C_z^T C_z & PE \\ E^T P & -\gamma^2 I \end{bmatrix} \prec 0. \quad (66)$$

The inequality (66) is nonlinear in the decision variables because (55) contains the product between K and P . To obtain a convex formulation, introduce the change of variables

$$X = P^{-1}, \quad Y = KX. \quad (67)$$

Applying a congruence transformation with $\text{diag}(X, I)$ gives

$$\begin{bmatrix} AX + BY + (AX + BY)^T & E \\ E^T & -\gamma^2 I \end{bmatrix} \prec 0.$$

The term $(C_z X)^T C_z X$ is quadratic in X . Using the Schur complement, (C) can be written as

$$\begin{bmatrix} AX + BY + (AX + BY)^T & E & (C_z X)^T \\ E^T & -\gamma^2 I & 0 \\ C_z X & 0 & -I \end{bmatrix} \prec 0. \quad (68)$$

This condition is linear in the decision variables X , Y , and γ^2 . Once a feasible solution is obtained, the feedback gain is recovered as (15). Thus, the resulting controller guarantees closed-loop stability while bounding the disturbance-to-performance-output gain. ■

D. Proof of Proposition 4

Proof: The measured data matrices satisfy the nominal data relation (5). Under the rank condition (7), the matrix contains sufficient information to recover a discrete-time representation of the system.

Thus, the estimated closed-loop system under disturbance is (20). Let's use (59). Then the disturbed closed-loop system becomes (16).

To impose dissipativity, the performance output is selected as (17). The goal is to bound the disturbance-to-performance-output gain from $w(k)$ to $z(k)$. A discrete-time bounded-real dissipativity condition can be written as

$$\begin{bmatrix} X & RX & E & 0 \\ (RX)^T & X & 0 & XC_z^T \\ E^T & 0 & \gamma I & 0 \\ 0 & C_z X & 0 & \gamma I \end{bmatrix} \succ 0. \quad (69)$$

However, this condition is nonlinear because $R = \hat{A} + \hat{B}K$ contains the product of the controller gain K and the Lyapunov variable X . To obtain a convex LMI, introduce the change of variables (67). Then

$$RX = (\hat{A} + \hat{B}K)X = \hat{A}X + \hat{B}KX = \hat{A}X + \hat{B}L. \quad (70)$$

Substituting this expression into (69) gives

$$\begin{bmatrix} X & \hat{A}X + \hat{B}L & E & 0 \\ (\hat{A}X + \hat{B}L)^T & X & 0 & XC_z^T \\ E^T & 0 & \gamma I & 0 \\ 0 & C_z X & 0 & \gamma I \end{bmatrix} \succ 0. \quad (71)$$

This LMI is linear in the decision variables X , Y , and γ , and can therefore be solved using semidefinite programming.

Once a feasible solution is obtained, the feedback gain is recovered as

$$K = YX^{-1}. \quad (72)$$

Therefore, the implemented controller (18), stabilizes the estimated closed-loop system while enforcing the dissipativity-based disturbance attenuation condition. ■

E. Proof of Proposition 5

Proof: Under the state-feedback law (18), the closed-loop dynamics are (23).

To establish dissipativity, consider the quadratic storage function (12). The closed-loop system is dissipative with respect to the supply rate if

$$V(x(k+1)) - V(x(k)) \leq \gamma^2 w(k)^T w(k) - z(k)^T z(k). \quad (73)$$

Using (23) and (17), the dissipation inequality becomes

$$(Rx + w)^T P(Rx + w) - x^T Px \leq \gamma^2 w^T w - x^T C_z^T C_z x.$$

Rearranging terms gives

$$\begin{bmatrix} x \\ w \end{bmatrix}^T \left(\begin{bmatrix} R^T PR - P + C_z^T C_z & R^T P \\ PR & P - \gamma^2 I \end{bmatrix} \right) \begin{bmatrix} x \\ w \end{bmatrix} \leq 0. \quad (74)$$

Equivalently, using the Schur complement, the above condition can be written in the lifted form

$$\begin{bmatrix} P^{-1} & 0 & R & I \\ 0 & I & C_z & 0 \\ R^T & C_z^T & P & 0 \\ I & 0 & 0 & \gamma^2 I \end{bmatrix} \succeq 0. \quad (75)$$

Introduce the change of variables

$$\bar{P} = P^{-1}, \quad \bar{K} = K\bar{P}.$$

Then

$$R\bar{P} = (A + BK)\bar{P} = A\bar{P} + B\bar{K}. \quad (76)$$

Applying the congruence transformation induced by $\bar{P}$ gives

the model-based dissipativity LMI

$$\begin{bmatrix} \bar{P} & 0 & A\bar{P} + B\bar{K} & I \\ 0 & I & C_z\bar{P} & 0 \\ (A\bar{P} + B\bar{K})^T & (C_z\bar{P})^T & \bar{P} & 0 \\ I & 0 & 0 & \gamma^2 I \end{bmatrix} \succeq 0. \quad (77)$$

The corresponding controller is recovered as (30).

Next, the model-based LMI is converted into a data-driven condition. From the measured data equation (24),

$$W_0 = X_1 - AX_0 - BU_0.$$

Using the disturbance bound (26). This condition can be written as

$$\begin{bmatrix} I \\ W_0^T \end{bmatrix}^T \begin{bmatrix} \delta_w I & 0 \\ 0 & -I \end{bmatrix} \begin{bmatrix} I \\ W_0^T \end{bmatrix} \succeq 0. \quad (78)$$

Since

$$W_0^T = \begin{bmatrix} X_1^T & -X_0^T & -U_0^T \end{bmatrix} \begin{bmatrix} I \\ A^T \\ B^T \end{bmatrix},$$

condition (78) can be written as

$$\begin{bmatrix} I \\ A^T \\ B^T \end{bmatrix}^T \Phi_w \begin{bmatrix} I \\ A^T \\ B^T \end{bmatrix} \succeq 0, \quad (79)$$

where Φ_w is defined in (27). Therefore, Φ_w characterizes all system matrices (A, B) that are consistent with the measured data and the disturbance bound.

The model-based LMI (77) must hold for all matrices (A, B) satisfying the data-consistency condition (79). By the matrix S-lemma, this implication is guaranteed if there exists a scalar $\lambda > 0$ such that the robust data-dependent LMI (28) holds, where $\bar{Q}_p$ is defined in (27). This eliminates the explicit dependence on the unknown matrices A and B and replaces it with the measured data matrices X_0 , X_1 , and U_0 (6), and the disturbance bound δ_w .

Finally, once a feasible solution $\bar{P}$ and $\bar{K}$ is obtained, the feedback gain is recovered as (30). Thus, the resulting controller is synthesized directly from data and guarantees dissipativity of the closed-loop system with respect to the supply rate (21). ■

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